

Stock Market Exploratory Data Analysis (1970–2018)

Project : Data Collection and Initial Analysis of Stock Market Data

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Software Used: Python, Pandas, Matplotlib, VS Code, Jupyter Notebook

Introduction

This project focuses on exploring historical stock market data collected between 1970 and 2018. The objective is to clean, prepare, and analyze stock market information to better understand how stock prices and trading activity evolved over time.

Two datasets were used throughout the project. The first dataset contains daily historical stock prices including Open, High, Low, Close, Adjusted Close, Volume, and Date. The second dataset contains company information such as ticker symbols, exchange names, company names, sectors, and industries.

The project follows a complete Exploratory Data Analysis (EDA) workflow beginning with data preparation and ending with statistical analysis and visualizations. The results provide a foundation for future predictive analytics and machine learning projects.

Methodology

The project followed these main stages:

- Data Collection
- Data Understanding
- Data Cleaning
- Data Segmentation by Decade
- Dataset Merging
- Exploratory Data Analysis
- Comparative Analysis
- Final Conclusions
- Future Hypotheses

Data preprocessing included handling missing values, removing duplicate records, converting the Date column into datetime format, and setting it as the DataFrame index. Company information was merged with stock price records using the ticker column.

Data Cleaning

The quality of the dataset was examined before performing any analysis.

The stock prices dataset contained no missing values. However, the company information dataset contained missing values within the **Sector** and **Industry** columns. Instead of deleting those records, missing values were replaced with "**Unknown**" to preserve all available companies while clearly identifying unavailable information.

No duplicate records were found in either dataset.

The Date column was converted into datetime format and used as the DataFrame index to support efficient time-series analysis.

Exploratory Data Analysis

The merged dataset was divided into five decades:

- 1970s
- 1980s
- 1990s
- 2000s
- 2010s

Summary statistics including the mean, median, and standard deviation were calculated for Open, High, Low, Close, and Volume.

Several visualizations were created to better understand market behavior across different decades.

Figure 1. Average Monthly Closing Prices by Decade

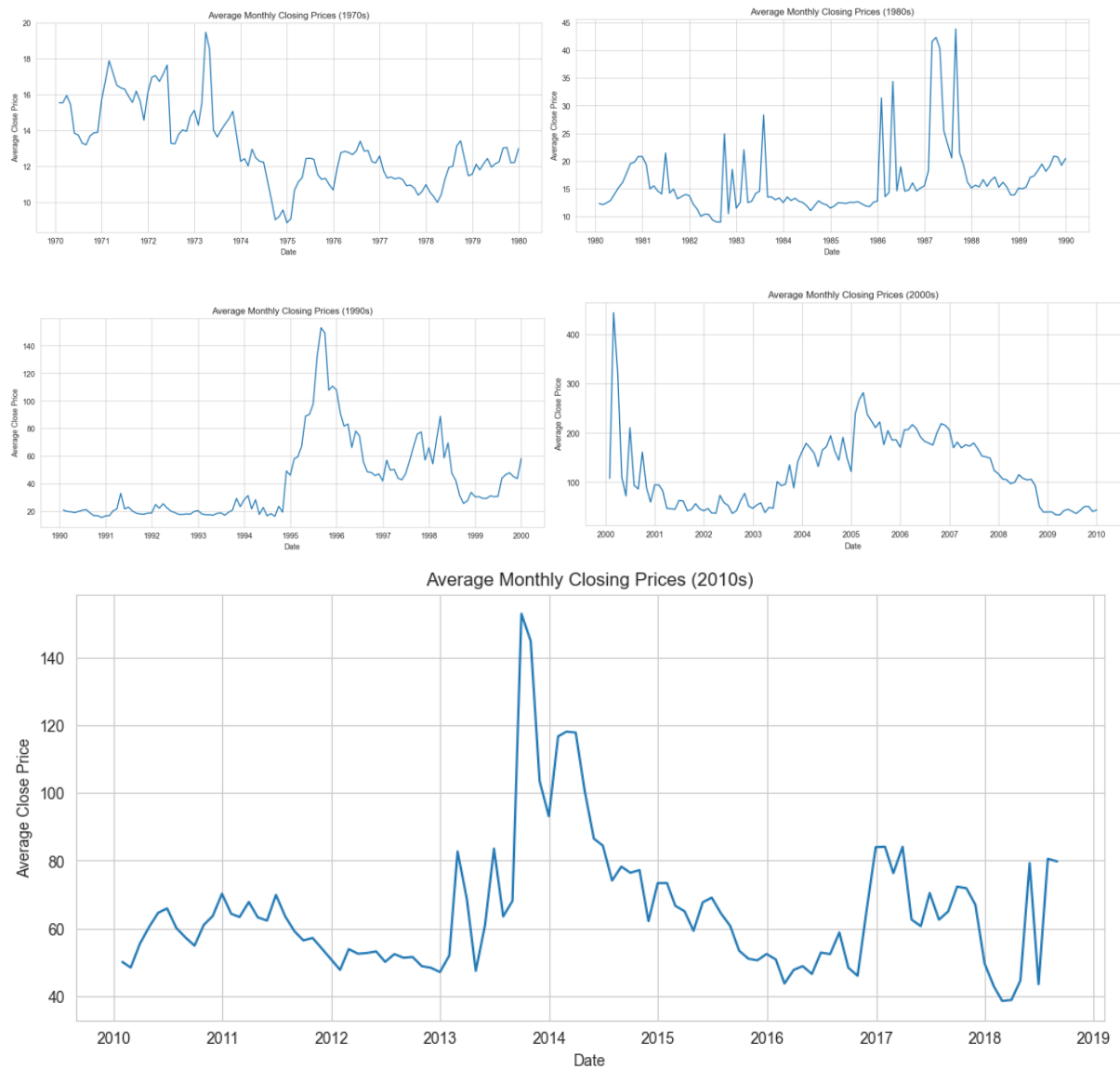


Figure 1 illustrates the monthly average closing prices across different decades. The visualizations show noticeable changes in stock market behavior over time, with later decades exhibiting stronger upward and downward price movements than earlier decades.

Figure 2. Trading Volume Distribution

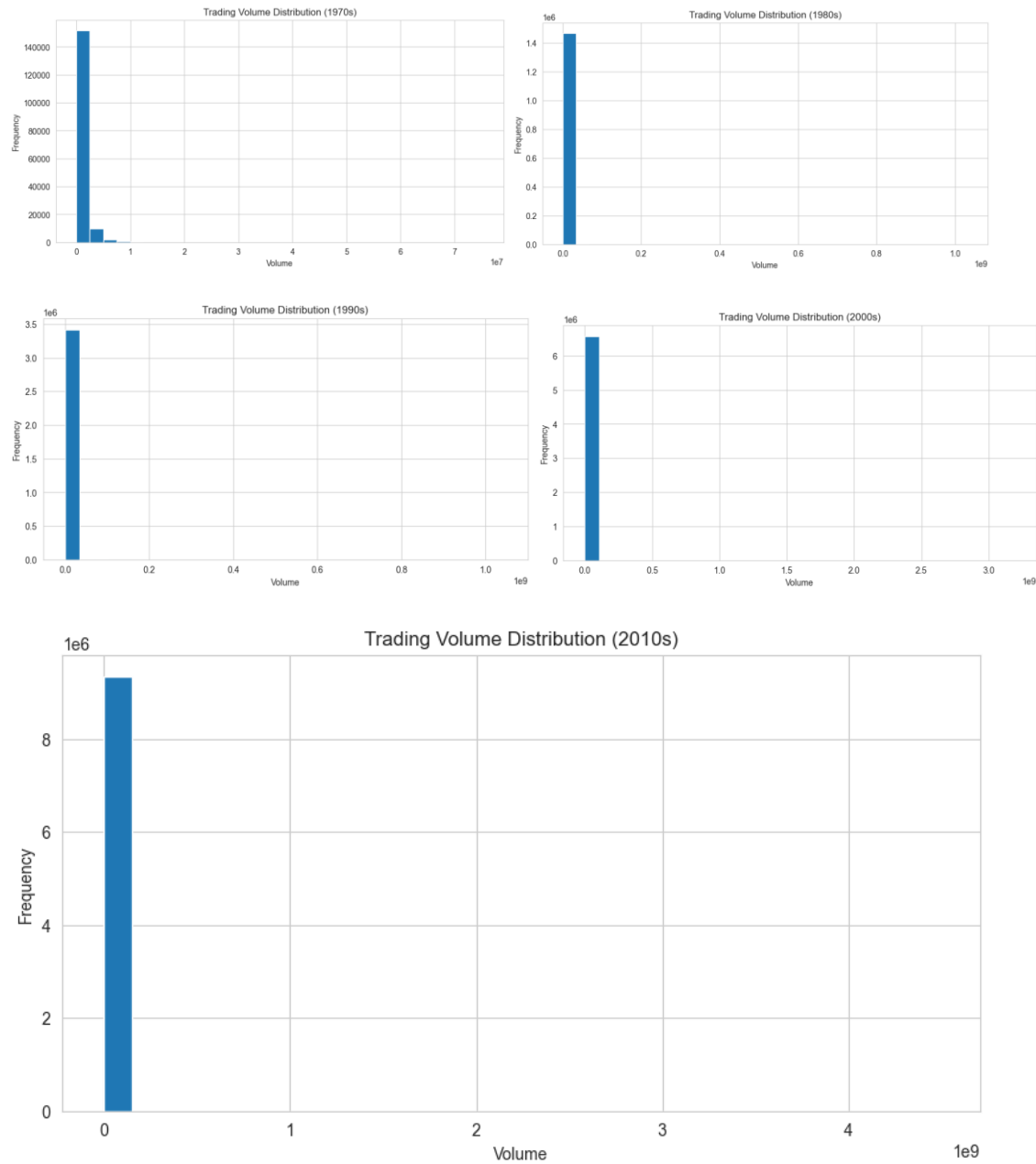


Figure 2 presents the distribution of trading volume for each decade. The histograms reveal highly right-skewed distributions, indicating that most trading days experienced moderate trading activity while relatively few observations recorded exceptionally high trading volumes.

Figure 3. High and Low Price Box Plots

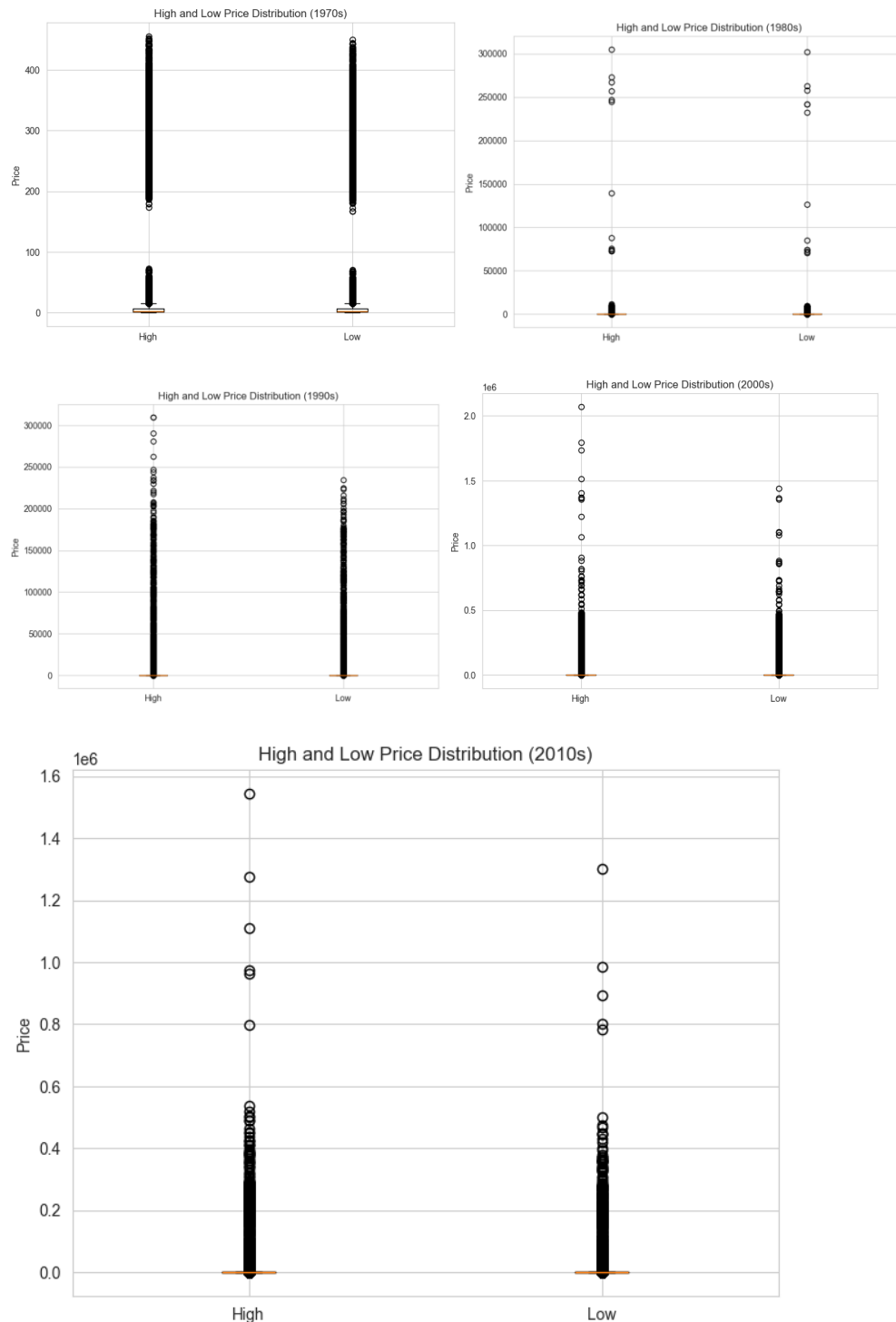


Figure 3 displays the distribution of High and Low stock prices across different decades. Numerous outliers are visible throughout the dataset, and the spread of prices generally becomes larger in later decades, indicating increased market variability.

5. Comparative Analysis

The statistical summaries and visualizations consistently demonstrated long-term changes in stock market behavior.

The 1970s showed relatively stable stock prices with moderate fluctuations, while the 1980s and 1990s experienced substantially greater volatility. The 2000s and 2010s continued to display large price movements and wider price distributions.

Trading volume increased considerably throughout the decades, suggesting greater market participation and increased investor activity. The higher standard deviation values observed in later decades confirmed that market volatility increased over time.

The histogram distributions demonstrated strong positive skewness, while the box plots identified numerous outliers across every decade. These findings consistently supported the conclusions obtained from the summary statistics

Conclusions

The exploratory data analysis successfully identified long-term trends within historical stock market data.

The datasets were successfully cleaned and prepared for analysis with no duplicate records and appropriately handled missing values.

The analysis revealed that:

- Stock prices changed significantly across different decades.
- Market volatility generally increased over time.
- Trading volume increased substantially in later decades.
- Multiple visualizations consistently confirmed the statistical findings.

Overall, the exploratory data analysis provided a clear understanding of historical stock market behavior and established a strong foundation for future predictive analytics projects.

Future Hypotheses

Based on the results of this exploratory analysis, several hypotheses could be investigated in future work:

- Companies from different sectors may exhibit different long-term price trends.
- Higher trading volume may be associated with greater price volatility.
- Certain industries may outperform others during specific economic periods.
- Historical stock prices could be used to develop predictive machine learning models.
- Seasonal factors may influence stock prices during particular months or years.

References

Pandas Development Team. *Pandas Documentation*.

Matplotlib Development Team. *Matplotlib Documentation*.

Historical Stock Prices Dataset (Course Dataset).